

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

CPU & Control Unit

Lecture No.- 2

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Recap of Previous Lecture



Topic

CPU

Topic

MIPS

Topic

Data Path

Topics to be Covered



Topic

Data Path

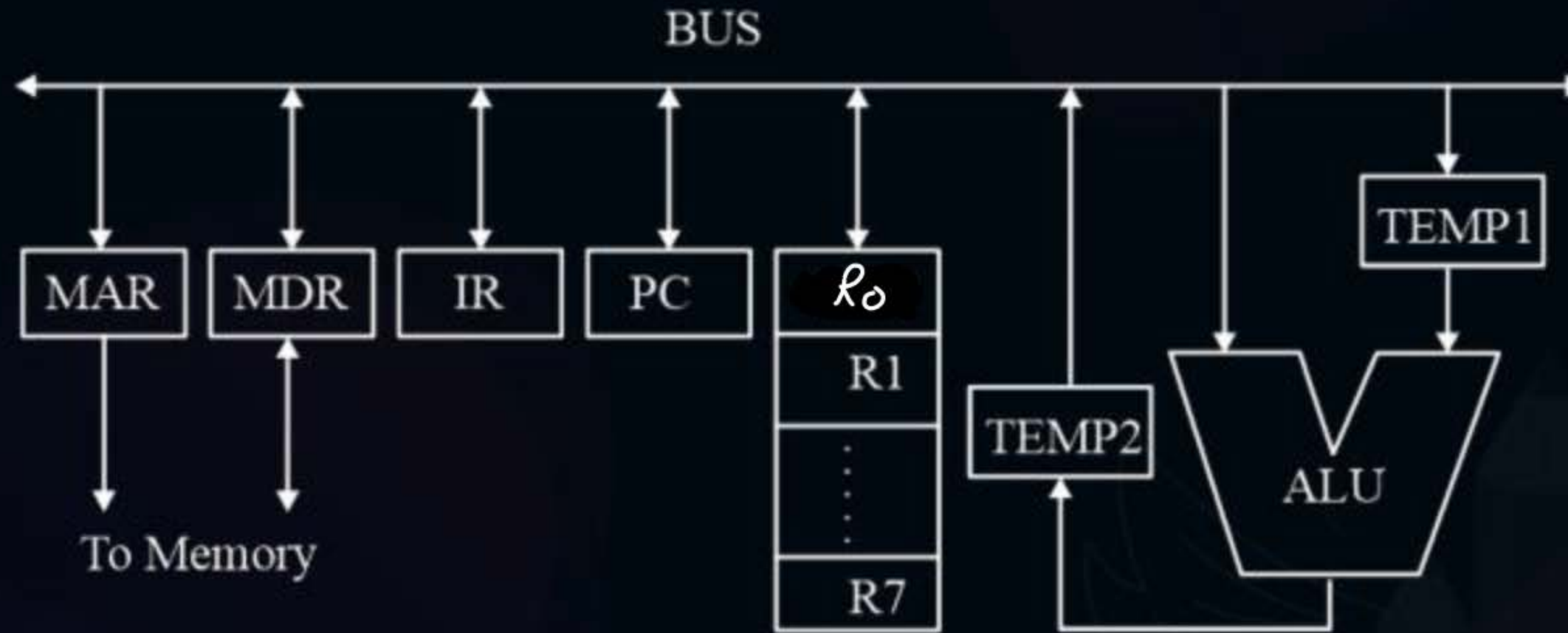
Topic

Control Unit

Topic

Hardwired Control Unit

#Q. Consider the following data path diagram



Consider an instruction: $R0 \leftarrow R1 + R2$. The following steps are used to execute it over the given data path. Assume that PC is incremented appropriately. The subscripts r and w indicate read and write operations, respectively.

1. $R2_r, TEMP1_r, ALU_{add}, TEMP2_w$
2. $R1_r, TEMP1_w$
3. PC_r, MAR_w, MEM_r
4. $TEMP2_r, R0_w$
5. MDR_r, IR_w

Which one of the following is the correct order of execution of the above steps

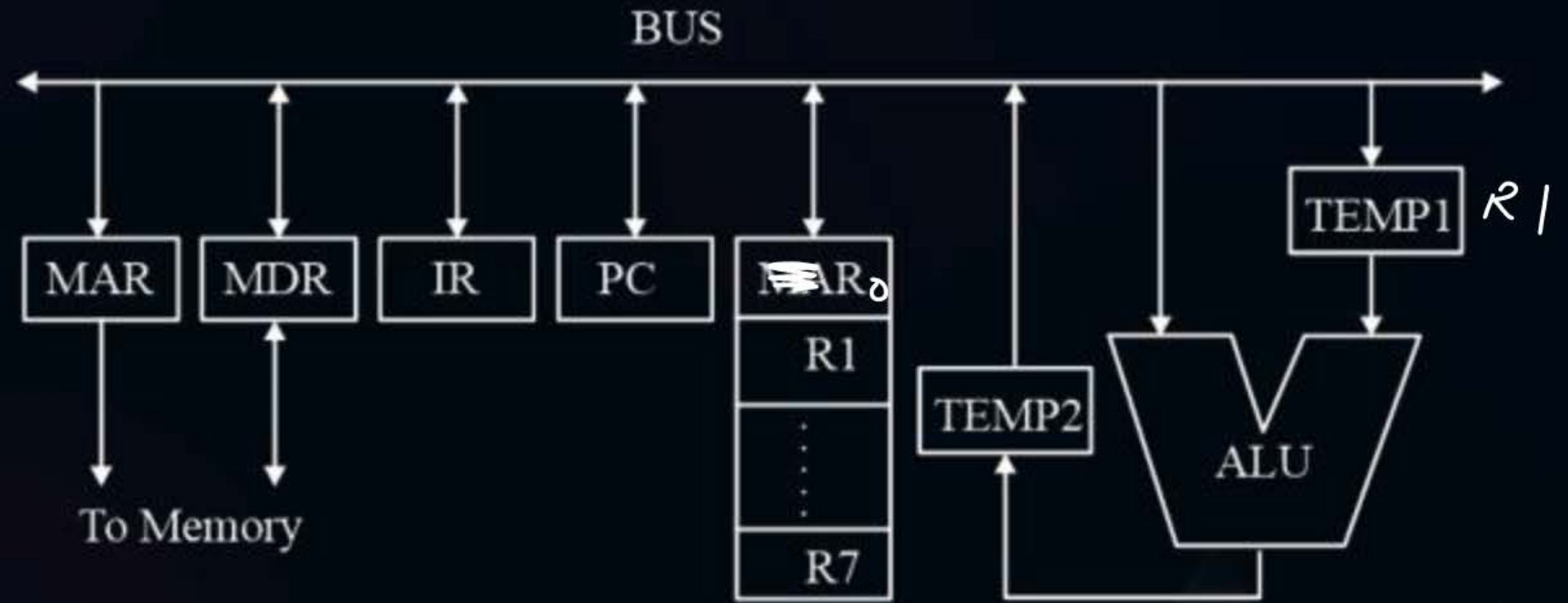
A 2, 1, 4, 5, 3

B 1, 2, 4, 3, 5

C ✓ 3, 5, 2, 1, 4

D 3, 5, 1, 2, 4

$R0 \leftarrow R1 + R2$

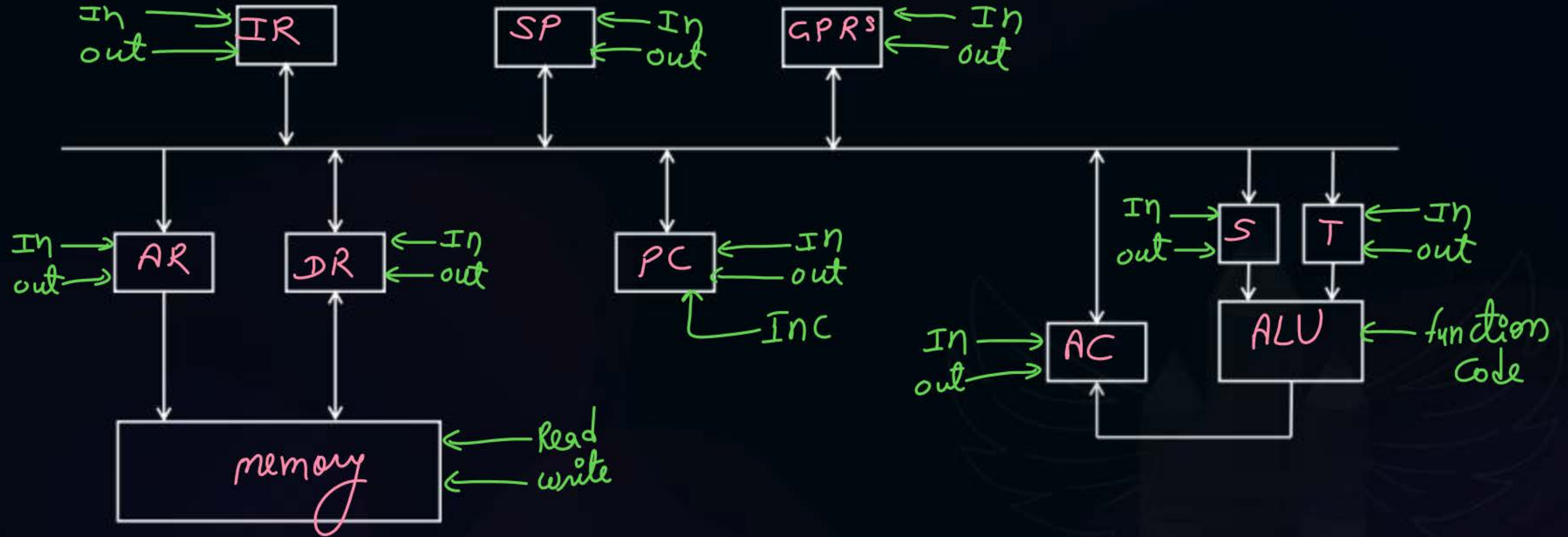


1. $R2_r, TEMP1_r, ALU_{add}, TEMP2_w$
2. $R1_r, TEMP1_w$
3. PC_r, MAR_w, MEM_r
4. $TEMP2_r, R0_w$
5. MDR_r, IR_w

3, 5, 2, 1, 4



Topic : Datapath





Topic : Control Unit

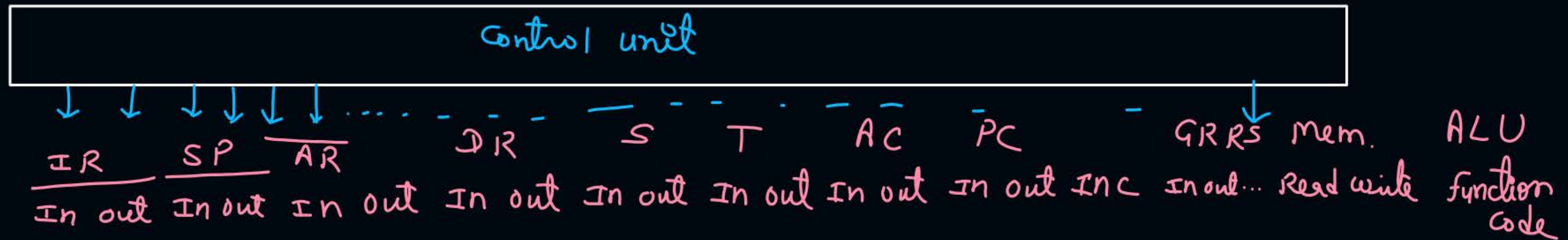


Control variable :-

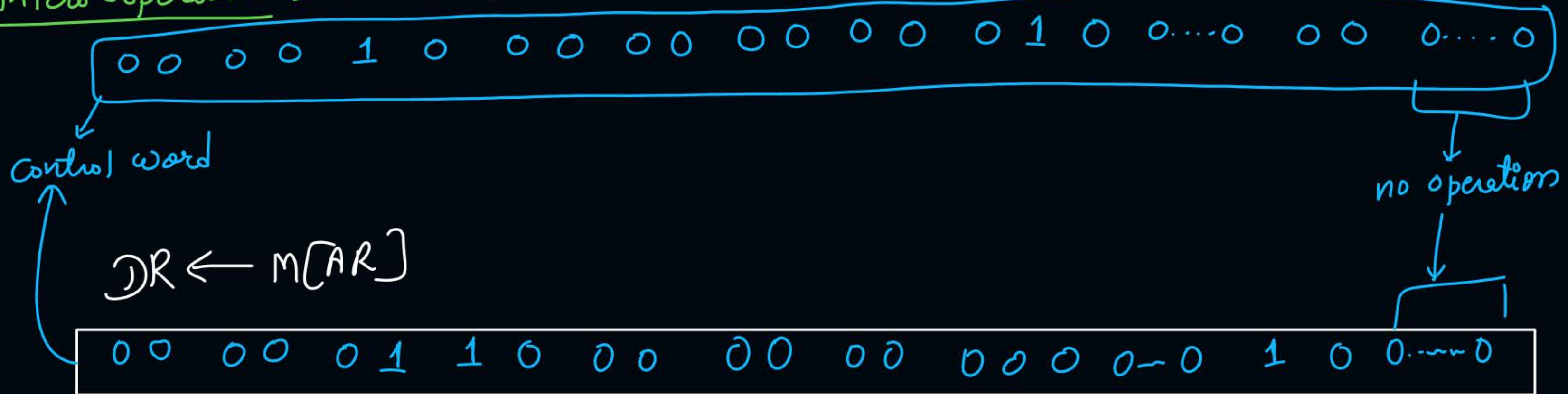
Name of a control signal

Control word :-

Collection of all control signals generated by control unit at a time.



micro-operation $\Rightarrow AR \leftarrow PC$



$IR \leftarrow DR, \quad PC \leftarrow PC + 1$

1	0	00	00	01	00	00	00	001	00---	00	00---
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Topic : Control Unit Organization

How control unit is designed to generate control words

1. Hardwired Control unit
2. Microprogrammed —||—



Topic : Hardwired Control Unit

Control logic is implemented with Gates, flip-flops, decoders and other digital circuits.

Advantage: Can be optimized to produce a faster mode of operation.

Disadvantage:

1. Rearranging the wires among various components is difficult.
2. Updation of control logic is difficult
3. Can be implemented for only simple computers

→ RISC

A CPU supports 4 distinct inst^{ns}.

$I_0 \Rightarrow \text{micro-op}^h \begin{matrix} 1 \\ 2 \\ \vdots \end{matrix} \Rightarrow \begin{matrix} \text{cw1} \\ \text{cw2} \\ \text{cw3} \\ \vdots \end{matrix}$

$S_0 \ S_1 S_2 S_3 S_4 \dots$
0001000100000---0
10000100100000---0

$I_1 \Rightarrow \text{micro-op}^h \begin{matrix} 1 \\ 2 \\ \vdots \end{matrix}$

I_2

I_3

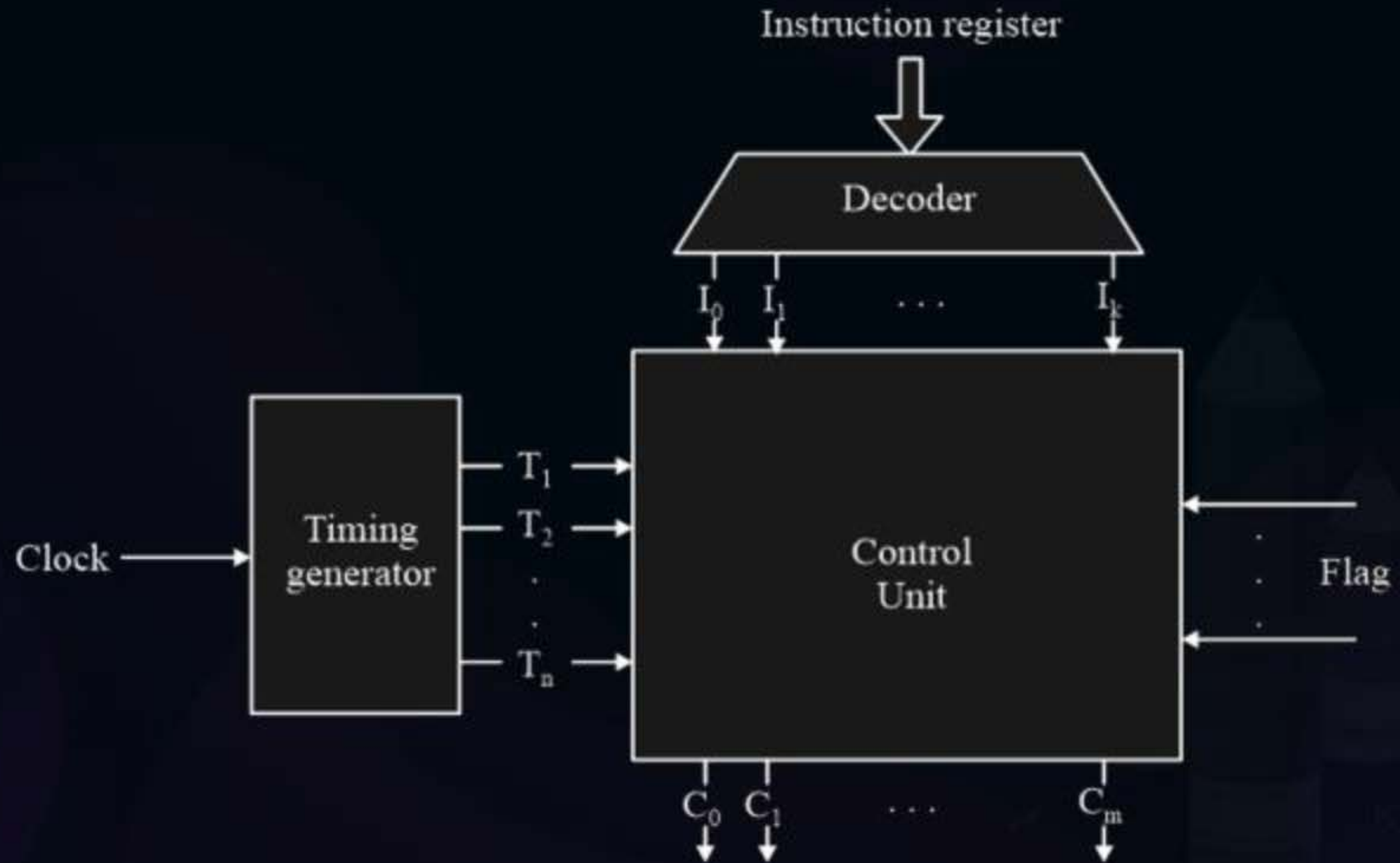
Table

Inst^{ns}

	cycles			
	T ₁	T ₂	T ₃	T ₄
I ₀	S ₃ , S ₇ ⇒ 1 remaining all 0	S ₀ , S ₅ , S ₈	—	—
I ₁				
I ₂				
I ₃				



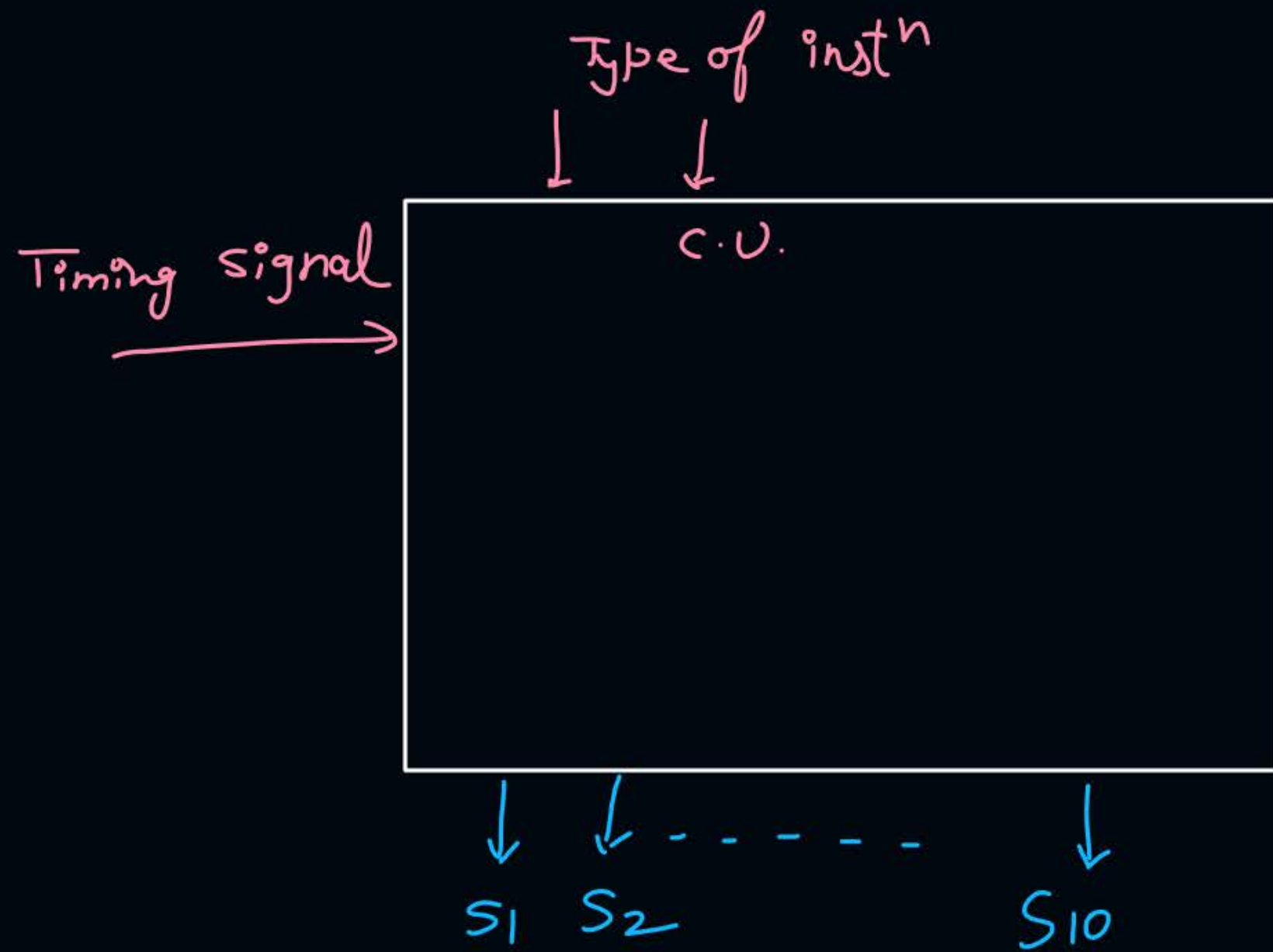
Topic : Hardwired Control Unit



#Q. A hardwired CPU uses 10 control signals S1 to S10, in various time steps T1 to T5, to implement 4 instructions I1 to I4 as shown below:

	T1	T2	T3	T4	T5
I1	S1, S3, S5	S2, S4, S6	S1, S7	S10	S3, S8
I2	S1, S3, S5	S8, S9, S10	S5, S6, S7	S6	S10
I3	S1, S3, S5	S7, S8, S10	S2, S6, S9	S10	S1, S3
I4	S1, S3, S5	S2, S6, S7	S5, S10	S6, S9	S10

Which of the following pairs of expressions represent the circuit for generating control signals S5 and S10 respectively?



$$S_5 = I_1 T_1 + I_2 T_1 + I_3 T_1 + I_4 T_1 + I_2 T_3 + I_4 T_3$$

$$= (I_1 + I_2 + I_3 + I_4) T_1 + (I_2 + I_4) T_3$$

$$= T_1 + (I_2 + I_4) T_3$$

A

$$S5 = T1 + I2 \cdot T3 \text{ and}$$

$$S10 = (I1 + I3) \cdot T4 + (I2 + I4) \cdot T5$$

B

$$S5 = T1 + (I2 + I4) \cdot T3 \text{ and}$$

$$S10 = (I1 + I3) \cdot T4 + (I2 + I4) \cdot T5$$

C

$$S5 = T1 + (I2 + I4) \cdot T3 \text{ and}$$

$$S10 = (I1 + I3 + I4) \cdot T2 + (I2 + I3) \cdot T4 + (I2 + I4) \cdot T5$$

D

$$S5 = T1 + (I2 + I4) \cdot T3 \text{ and}$$

$$S10 = (I2 + I3) \cdot T2 + I4 \cdot T3 + (I1 + I3) \cdot T4 + (I2 + I4) \cdot T5$$



Topic : Micro-Programmed Control Unit

all possible control words are stored in a memory inside control unit.

Control logic is implemented with micro-programs.

Advantage:

1. Updating the control logic is easy.
2. Easier to implement for complex computers also

Disadvantage: Slower than hardwired control unit.



2 mins Summary



Topic

Data Path

Topic

Control Unit

Topic

Hardwired Control Unit



Happy Learning

THANK - YOU

